
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Despite their remarkable overall fidelity, generative image models trained on large-scale datasets systematically fail to render seemingly elementary concepts—e.g., human hands or objects appearing in groups of four—that are amply represented in their training data. Such failures have so far been catalogued only anecdotally, leaving unresolved whether they constitute isolated idiosyncrasies or the surface manifestations of a structural limitation shared across models. We resolve this question with a systematic, unsupervised methodology for identifying and characterizing “conceptual blindspots”—concepts that are present in the training distribution yet absent from, or distorted in, a model’s generations. Our method leverages sparse autoencoders (SAEs) to extract interpretable concept embeddings, which we use to quantitatively compare the prevalence of each concept between real and generated images. As the representational backbone of this comparison, we train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE reported to date—affording concept-level resolution of the disparities between a model’s output distribution and the natural image distribution. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), our approach pinpoints concrete suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated blindspots (e.g., wood background texture and palm trees). Descending to the level of individual datapoints, we further isolate memorization artifacts—instances in which models reproduce highly specific visual templates encountered during training. Together, these results establish a theoretically grounded framework that diagnoses generative models by measuring the conceptual fidelity of their outputs against the underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have attained unprecedented capabilities, driving applications both within the vision domain [1–9] and well beyond it [9–15]. This success notwithstanding, a growing body of qualitative [16–18] and quantitative [19–20] evidence documents that these models routinely fail to render even elementary concepts—among them human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. Tellingly, when prompted for such concepts, models do not omit them outright but instead synthesize closely related yet incorrect structure—for instance, rendering hands with six fingers. We term these failure modes “conceptual blindspots”¹ and their existence is counterintuitive precisely because the training corpora contain abundant, faithful demonstrations of the very concepts that are missed. This tension motivates the central question of our work: are

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ quantifies a model’s tendency to over- or under-generate a concept c_k relative to its frequency in the natural data. Concepts with $\delta(k) < 0.1$ are designated **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ **exaggerated conceptual blindspots**. The images shown, produced by four widely used generative image models, contrast concepts that fall into each blindspot regime against concepts the models render faithfully.

such failures the idiosyncratic quirks of a handful of concepts, or the signature of a systematic phenomenon in which, across a broad spectrum of concepts, models are consistently either unable or overly inclined to generate them?

Existing approaches for evaluating generative image models are structurally ill-suited to answering this question [28, 29]. The prevailing metrics are coarse-grained scores of image realism—chief among them FID [30]—which aggregate over entire images and are therefore blind to the concept-level distributional failures at issue. Diversity- and coverage-oriented metrics such as CLIPScore [31–33] move closer, but they operate at the granularity of whole images or prompts rather than that of individual features or concepts [34, 35], and thus cannot localize which concepts a model fails to reproduce. Qualitative protocols—participant surveys [5, 17, 36–38] and open-ended exploration [39]—can surface individual conceptual failures but do not constitute a scalable, repeatable methodology that transfers across model families and supports systematic comparison.

This work. We address this gap by developing an automated, unsupervised methodology that identifies concepts in the data distribution whose odds of generation under the trained model diverge from their odds under the true data-generating process. Crucially, rather than merely quantifying distribution coverage, the methodology pinpoints *which* concepts constitute the mismatch and whether each is suppressed or exaggerated. Our contributions are three-fold:

- **A formalization of conceptual blindspots in generative image models.** We cast conceptual blindspots as measurable discrepancies between the concept content of natural images and that of a model’s generations, and derive a principled, quantitative measure of these discrepancies (Section 3). This formalization supplants anecdotal and human-defined concept evaluations with a criterion grounded in the data-generating process itself.
- **A scalable, unsupervised method for detecting conceptual blindspots with sparse autoencoders.** We operationalize the formalization as a concrete pipeline that uses sparse autoencoders (SAEs) to extract and compare concept distributions between natural and synthesized images (Section 4). To this end, we train and open-source an archetypal SAE (RA-SAE) on DINOv2 features spanning 32,000 concepts—the largest such RA-SAE reported to date—which supplies the concept-level resolution the analysis requires.
- **An exploratory tool and a multi-level empirical analysis.** We release an interactive web tool that supports both distribution- and datapoint-level analysis of blindspots across models (Sections 5.1–5.4), and apply our method to four widely used models—Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5). The analysis surfaces concrete suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), illustrated in Fig. 1.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early work in explainable AI, computer vision included, centered on attributing predictions to influential input regions [40, 42]. Such attribution methods, however, convey little semantic information about the learned representations and can yield explanations that are demonstrably unfaithful [43–45].

76 Concept-based interpretability [46] arose to overcome these shortcomings by identifying semantically
77 meaningful directions in a network’s representation space, thereby revealing not merely where a
78 model attends but which concepts and structures it encodes [47–49].

79 Recent work establishes that the leading concept-based interpretability methods—ACE [44],
80 CRAFT [48], and sparse autoencoders (SAEs) [50–51]—all instantiate the same underlying dictionary-
81 learning problem under differing constraints [52]. Among them, SAEs have proven the most scalable
82 for concept-based interpretability. Although the original formulation exhibits known limitations—
83 overly specific features [53], compositionality challenges [54], and limited intervention effects [55]—
84 subsequent refinements address them directly, notably the archetypal SAE (RA-SAE) [56] that we
85 adopt here.

86 Complementary interpretability techniques—prompt-based probing [57], attention-map and activa-
87 tion visualizations [58–59], and dataset-level statistics such as diversity or distribution-coverage
88 metrics [32, 33]—each afford only partial purchase on our question. Their common limitation is
89 granularity: they operate over whole images or prompts rather than the fine-grained features and
90 concepts that blindspot analysis demands [34–35]. They further rely on subjective interpretation
91 and do not disambiguate distinct failure modes [60]. Existing methods and metrics are, in short,
92 inadequate for systematically localizing the feature- and concept-level weaknesses of generative
93 image models [28]—the gap our framework is designed to close.

94 2.2 Generative Image Models

95 Diffusion-based methods now dominate generative vision modeling across modalities, spanning
96 image [4, 61–64], video [65–72], and 3D [7, 73–75] synthesis. In image generation, the lineage
97 traces to denoising diffusion probabilistic models (DDPMs) [76], subsequently generalized to non-
98 Markovian diffusion processes by denoising diffusion implicit models (DDIMs) [77].

99 The Stable Diffusion (SD) [9] family brought DDPMs to both the research and open-source commu-
100 nities at scale, and was extended through successive releases—SD 2 [78], SD 3 [79], and SD XL [80].
101 A large ecosystem of architectural modifications followed, imposing additional constraints on the
102 diffusion process—for example on style [81–82], pose [83], and identity [84–85]—and admitting
103 alternative input modalities such as image-to-image generation. A parallel line of latent diffusion mod-
104 els built atop SD—Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89]—has likewise
105 proliferated, and it is precisely this diversity of architectures that our model-agnostic methodology is
106 built to compare.

107 2.3 Datasets for Generative Image Models

108 The recent success of generative vision models rests largely on the confluence of abundant com-
109 putation and large-scale internet datasets [90–92]. LAION-5B [93] has been pivotal to training
110 open-source text-to-image models, SD and its derivatives among them. Scraped from Common
111 Crawl [94], it comprises over 5B image-caption pairs, 2.3B of them in English; other prominent cor-
112 pora include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M pairs, respectively.

113 The dataset’s prominence has been accompanied by mounting concerns over its biases [97–101].
114 Filtering efforts notwithstanding, harmful content persisted [97–99]—including NSFW material [100]
115 and hundreds of CSAM instances [101]—prompting the corpus’s temporary withdrawal from official
116 channels. The data is further marked by low-quality images [102] and internet-style captions (e.g.,
117 product descriptions) that diverge from how users actually prompt the trained models [103]. These
118 distributional properties directly shape the conceptual blindspots our analysis is designed to expose.

119 2.4 Concept Discovery and Sparse Coding in Generative Image Models

120 Dictionary learning seeks sparse representations in which each sample is reconstructed as a linear
121 combination of a few dictionary atoms [104–106]. Grounded in compressed sensing theory [107–108],
122 the field has progressed from early vector quantization [109] to a rich family of formulations, including
123 non-negative matrix factorization [110–111], sparse PCA [112], and K-SVD [113]. More recent
124 developments encompass online algorithms [114], structured sparsity [115], and theoretical recovery
125 guarantees [116–117], together with increasingly tight connections to deep learning [118–119].

Sparse coding has, in turn, been brought to bear on the emergence of high-level concepts inside diffusion models [120]. Before diffusion models, concept-grounded interpretability was applied to earlier generative architectures via concept-bottleneck models, which demand human intervention at training time [121], and post-hoc detectors that retrofit concept supervision [122]. Both paradigms, however, presuppose a fixed set of human-defined concepts and therefore cannot register broader trends the user has not anticipated—the very trends an unsupervised method can surface.

To evaluate generative image models, aggregate metrics—precision, recall, density, and coverage [35] [123]—and latent density scores that predict sample quality from the model’s latent space [124] have been proposed. These reliably expose distributional gaps in aggregate, yet remain silent on which specific concepts are under- or over-represented.

Taken together, these limitations motivate a scalable, unsupervised framework that systematically identifies and quantifies concept-level failure modes in generative image models [125]—the framework we develop next.

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing *conceptual blindspots* as systematic discrepancies between the conceptual content of natural images and that of a model’s generations. This formulation yields principled, quantitative measures that identify precisely which concepts a generative model under- or over-represents relative to its data distribution; the overall process is illustrated in Fig. 2. While the derivation invokes standard assumptions from the identifiability and disentanglement literature [126–130], we find empirically that the framework elicits meaningful phenomenology even when those assumptions are violated, indicating that its diagnostic value does not hinge on their exact satisfaction.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational economy, we let $p(\cdot)$ denote both the latent density $p(c)$ and its pushforward to image space $p(x)$, where $x = G(c)$. The invertibility of G legitimizes this overloading: it induces a well-defined distribution over $\mathcal{X}$ through the change-of-variables formula. The individual dimensions of the latent space thus constitute the **Concepts** underlying the data distribution $P_{\mathcal{X}}$, defined over the observation space of images $\mathcal{X}$. Concretely, distinct latents may encode concepts such as color, shape, size, and location [131] [132]. We additionally let the data-generating process associate a text description $t \in \mathcal{T}$ with each sampled image, though we do not model this association explicitly. These descriptions serve as the conditioning signal for a text-conditioned **Generative image model** g_{θ} with parameters θ : trained on a corpus of image-text pairs, g_{θ} maps a noise signal $\eta \sim \mathcal{N}(0, I)$ together with a scene description t to an image x realizing that description.

Defining conceptual blindspots in g_{θ} therefore reduces to comparing the probability mass a concept receives under the data-generating process with the mass it receives under the model. To operationalize this comparison, we take an evaluation set of natural image-text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and construct a matched set of synthesized images $\mathcal{D}'_{\mathcal{X}}$ from the same text descriptions. We estimate a concept’s probability of occurrence through an **Energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to a d -dimensional vector whose k^{th} coordinate is the scalar energy of the k^{th} concept, i.e., $\xi_k(x) = E(c_k)$. The invertibility of the data-generating process is what renders these estimates recoverable. The probability the process assigns to an image

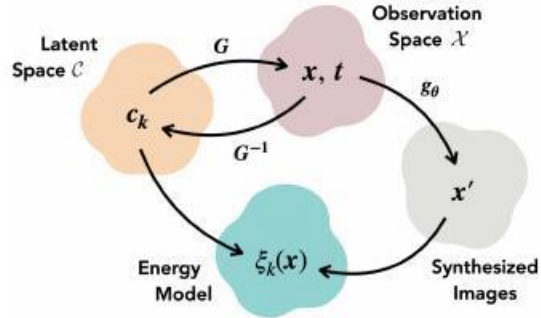


Figure 2: Schematic of our framework. Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) by the invertible DGP. Conditioned on the caption t , S_{θ} synthesizes an image x' . The energy model $\xi_k(x)$ recovers per-concept representations from both x and x' , so that comparing their concept distributions localizes blindspots.

180 $\mathbf{x}$ then follows as $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (with Z_ξ the partition function), which aggregates to
 181 the population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{\mathbf{x} \in \mathcal{D}_\mathcal{X}} p_k(\mathbf{x})$. Combining this with the sigmoid map $\sigma(\cdot)$,
 182 we now define the average energy difference in the k^{th} concept between the datasets $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$.
 183 **Definition 2 (Energy Difference).** Let $\mathbf{x} \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-
 184 generating process G , and let $\mathbf{x}' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let
 185 $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the
 186 energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

187 where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
 188 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

189 The energy difference in the k^{th} concept thus expresses the relative probability that the concept occurs
 190 in a target set of observations (here, $\mathcal{D}'_\mathcal{X}$) versus a baseline set (here, $\mathcal{D}_\mathcal{X}$), normalized to $(0, 1)$ by the
 191 sigmoid so that 0.5 denotes parity. This measure furnishes a direct criterion for conceptual blindspots.

192 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
 193 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
 194 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

195 In sum, a conceptual blindspot is a concept whose likelihood of occurrence in generated images
 196 departs markedly—through either suppression or exaggeration—from its prevalence under the data-
 197 generating process. Suppressed concepts activate disproportionately rarely ($\delta(k) < \lambda_{\min}$), whereas
 198 exaggerated concepts are overrepresented ($\delta(k) > \lambda_{\max}$); throughout, we set $\lambda_{\min} = 0.1$ and
 199 $\lambda_{\max} = 0.9$ to demarcate these two regimes.

200 This definition refines the classical notion of “mode collapse” studied in past work (e.g., Bau et al.
 201 [39]), from which it differs chiefly in granularity. Whereas mode collapse concerns the suppressed or
 202 exaggerated odds of generating *entire images*, our measure isolates the shifted odds of *individual*
 203 *concepts*. Thus, for instance, a model’s failure to produce objects on a `white` background registers,
 204 in our formalism, as a suppressed conceptual blindspot for that specific concept rather than as a global
 205 collapse.

206 4 Method: Operationalizing the Definition of Conceptual Blindspots

207 We now instantiate the formalization of Sec. 3 as a concrete pipeline for identifying conceptual
 208 blindspots in a generative model g_θ . Two ingredients are required: (i) a set of images sampled from
 209 g_θ that can be placed in one-to-one correspondence with samples from the ground-truth generative
 210 process, and (ii) an energy model that renders the two sets comparable in a common concept basis.
 211 For notation, $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_0$ the number of non-zero entries (the ℓ_0
 212 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ denotes element-wise non-negativity. For $n > 0$, we
 213 write $[n] := \{1, \dots, n\}$ and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

214 **From Prompts to Latent Representations.** To identify conceptual blindspots in g_θ , we compare
 215 a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs $(\mathbf{x}, t)$ drawn from the data-generating process G against their
 216 synthetic counterparts, generated by g_θ from the identical text descriptions so that each real image is
 217 matched to a prompt-aligned generation.

218 Concretely, given t , we synthesize the counterpart image $\mathbf{x}'$ with a pretrained text-to-image generator
 219 $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, realized as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Sampling
 220 proceeds in latent space along the reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\alpha_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

221 where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The terminal latent
 222 γ_0 is decoded by a pretrained VAE to produce the synthetic image $\mathbf{x}' = \text{VAE}(\gamma_0)$. For the remainder
 223 of the paper we treat g_θ as a black box mapping prompts to images, $t \mapsto \mathbf{x}'$; this abstraction is

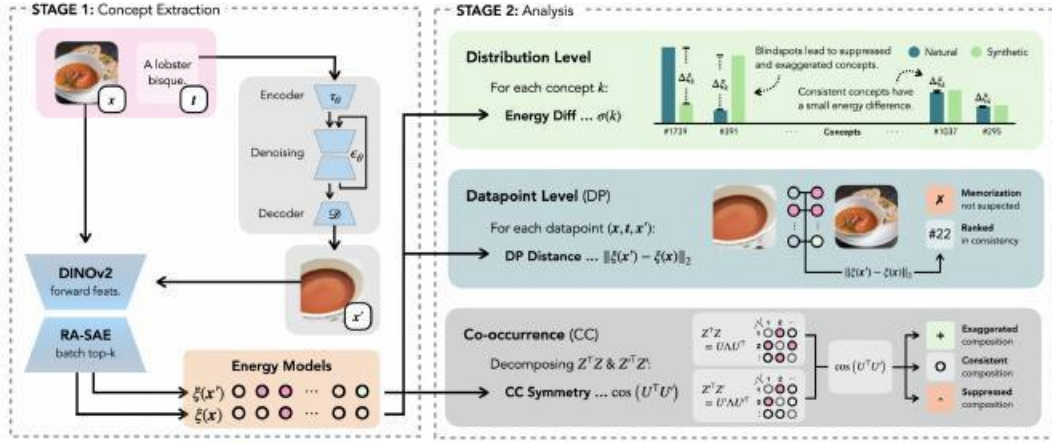


Figure 3: **Concept extraction pipeline.** For a triplet (x, t, x') , concepts are extracted by first computing the DINOv2 features of each image and then mapping them through the RA-SAE into sparse concept embeddings, which constitute the energy models $\xi(x)$ and $\xi(x')$. By construction, precisely what makes the method architecture-agnostic and applicable uniformly across the diffusion models we study.

Defining the Energy Model. Prior work establishes that self-supervised learning methods can invert the data-generating process and recover its underlying energy function up to a linear transformation [126, 128, 133, 134]; we exploit this property by adopting DINOv2 [135], a state-of-the-art self-supervised encoder, as the backbone of our energy model. Because the number of concepts underlying the DGP is expected to exceed the dimensionality of the encoder’s feature space [51, 136], these concepts are represented in superposition; we therefore train sparse autoencoders (SAEs) on the DINOv2 features to disentangle the subspaces that correspond to individual concepts [50, 56, 137–139]. The rationale is precise: if DINOv2 encodes the generative concepts along approximately orthogonal directions—consistent with the independence structure posited in Def. 1—then an SAE trained on its features will isolate each concept along a distinct latent dimension [136]. The activation the SAE assigns to that dimension then serves as our estimator of the ground-truth energy of the concept it encodes.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the feature extractor (the DINOv2 model); we compute features $a = f(x) \in \mathbb{R}^d$ for both the natural and synthetic images in $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$. Writing n for the number of image-text pairs, we stack the real and generated features into matrices $A, A' \in \mathbb{R}^{n \times d}$. Each feature vector is then decomposed into a sparse combination of learned concept atoms via an SAE. Let $D \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors and $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ the SAE encoder that maps features to sparse codes. Applying Ψ row-wise produces the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, whose i -th row $z_i = \Psi(a_i)$ is the concept decomposition of the i -th image. The decoder reconstructs features through ZD^T , and the SAE is trained to minimize reconstruction error subject to sparsity and non-negativity constraints:

$$\min_{\Psi, D} \|A - \Psi(A)D^T\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs are prone to converging on arbitrarily oriented dictionaries, rendering downstream analyses acutely sensitive to the random seed. To eliminate this instability and guarantee seed-independent, reproducible results, we employ the *Archetypal* SAE (RA-SAE) [56] under a TOP-K sparsity constraint [137]. RA-SAE restricts each dictionary atom to a convex combination of the training data. Formally, we write $D = WA$ with $W \in \Omega_{K', n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Consequently, every atom D_i lies within the convex hull of the data $\text{conv}(A)$, and any reconstruction ZD^T resides within the conic hull $\text{cone}(A)$. This constraint tethers the learned concepts to the support of the data distribution, ensuring they remain faithful to it [56]. Once trained, the SAE yields a single, consistent concept basis in which both real and generated images are expressed: Z for real images and Z' for their generated counterparts. *Because both sets of codes are expressed over the same interpretable concepts, they render prompt-conditioned visual semantics directly comparable,*

and each concept’s activation supplies the energy value on which our blindspot analysis operates. Our method therefore constitutes a structured pipeline that, from a prompt and its associated real image (t, x) , returns two sparse concept vectors (z, z') and thereby places the real and generated visual content in a common conceptual basis for direct comparison.

To summarize the full pipeline: beginning from a captioned image (t, x) , we synthesize a counterpart x' and embed both images into a shared sparse concept space through a vision encoder and the trained SAE, obtaining interpretable representations (z, z') that form the foundation for quantifying the conceptual shifts induced by the generative process.

5 Results

We analyze four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—over $|\mathcal{X}| = 10,000$ image-text pairs and their prompt-matched generations (Fig. 4). The analysis proceeds at three complementary levels (Fig. 5): a **distribution-level** evaluation reveals suppressed and exaggerated concepts; a **datapoint-level** analysis surfaces failures attributable to ambiguity, omission, and memorization; and a **compositional** analysis exposes structured distortions in the geometry of concept co-occurrence.

At the center of our contribution is an interactive exploratory tool, shown in Fig. 5, a web-based interface organized around a UMAP projection of concept representations that supports visualization and comparison of concept-level energy differences. It is publicly available at a software repository together with pre-computed energy-difference data for all four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky), and every analysis reported below is derived through it. Its principal capabilities are:

- **Contrasting models and architectures.** For each evaluated model, the tool renders a UMAP visualization spanning all 32,000 RA-SAE concepts, with every scatter point representing an individual concept color-coded by its energy difference.
- **Inspecting concepts.** Each concept is documented by a card reporting key statistics, its most representative real and generated images (x, x') , and its co-occurrence patterns.
- **Exploring blindspots.** Beyond the UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, foregrounding the most pronounced conceptual blindspots for inspection.

Rather than exhaust a single phenomenon, we report a set of high-level findings that together demonstrate the tool’s versatility and open the door to broader, user-directed exploration.

5.1 Generative Image Models Cannot Produce Unseen Concepts

To characterize the disparities between the generative models and the natural image distribution, we first analyze the marginal energy difference $\delta(k)$ across the 32,000 concepts learned by the RA-SAE. As defined in Sec. 4, this quantity measures the relative prevalence of each concept in the synthesized versus the natural image set: $\delta(k) < 0.1$ marks concept k as under-represented (suppressed) in the generated images, and $\delta(k) > 0.9$ marks it as over-represented (exaggerated). Fig. 6 presents the distribution of $\delta(k)$ for each of the four models. Every model exhibits a heavy-tailed histogram with substantial mass at both extremes, establishing that concept-coverage discrepancies are systematic rather than incidental. Decisively, the left tail—the suppressed concepts—is consistently denser and longer than the right, evidencing a robust and shared tendency toward concept suppression. This asymmetry is quantified by the uniformly negative skewness of the distributions: -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

While all four models thus display both suppressed and exaggerated concepts, their detailed profiles diverge. PixArt, for example, exhibits the widest spread, consistent with a more heavily suppressed concept distribution. The left skew common to every distribution nonetheless points to a shared



Figure 4: **Representative image-caption pairs** from LAION-5B, each shown alongside the images synthesized from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

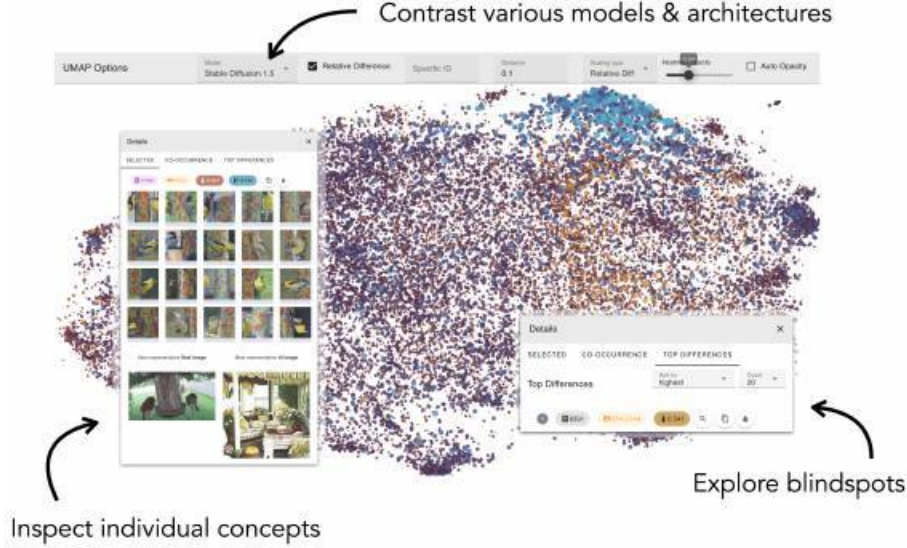


Figure 5: **Overview of the exploratory tool.** The web interface displays a per-model UMAP projection in which each dot is a concept color-coded by its energy difference. Selecting a concept opens a detail panel that reports its statistics and its most representative natural and generated images, x and x' , together with an ordered list of the concept’s co-occurrences and the global blindspot rankings.

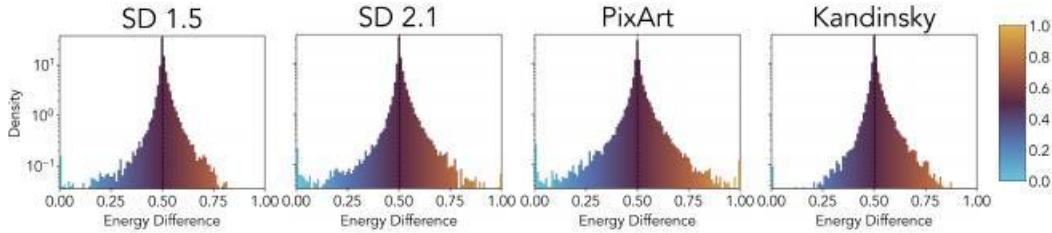


Figure 6: **Concept energy distribution.** Log-scale histograms of the energy difference $\delta(k)$ over all 32,000 concepts, contrasting the natural and synthesized distributions for each evaluated model. Mass to the left corresponds to suppressed (under-represented) concepts and mass to the right to exaggerated (over-represented) concepts; the pronounced left skew of every histogram signals a systematic bias toward suppression.

propensity for concept omission; the finer structure of this behavior is the subject of the sections that follow.

5.2 Structure and Specificity of Conceptual Blindspots

Whereas the previous section quantified the marginal discrepancies in concept frequency, we now characterize their global structure by embedding all 32,000 concepts into two dimensions via UMAP on the sparse codes, again coloring each concept by its $\delta(\cdot)$ value. As Fig. 7 shows, distinct concept clusters emerge for every model. These clusters frequently coincide with contiguous regions of conceptual blindspots—most conspicuously for suppressed (blue) concepts—demonstrating that blindspots are not randomly distributed but structured, and hence traceable to shared biases in the training distribution or to architectural priors. To determine how consistent these blindspot patterns are across models, we analyze both the magnitude and the structure of the concept-level $\delta(\cdot)$ values. Fig. 8 reports scatter plots and pairwise Pearson correlations between the $\delta(k)$ vector of SD 1.5 and those of the remaining models. SD 1.5 and SD 2.1 are strongly correlated ($r = 0.82$), as expected from their shared architecture and training pipeline. Their correlations with PixArt and Kandinsky, by contrast, are markedly lower ($r = 0.41$ and $r = 0.46$, respectively), establishing that these architectures allocate their blindspots to substantially different regions of the conceptual space.

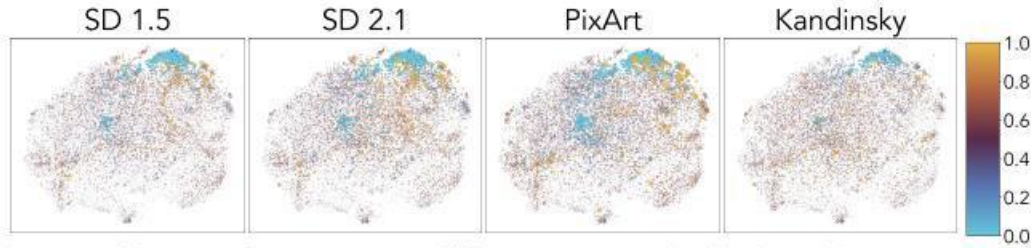


Figure 7: **Structure of concept energy differences.** UMAP visualizations of the 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions. Coherent clusters—rather than scattered points—reveal that conceptual blindspots are spatially organized, with suppressed concepts at the blue end and exaggerated ones at the yellow end.

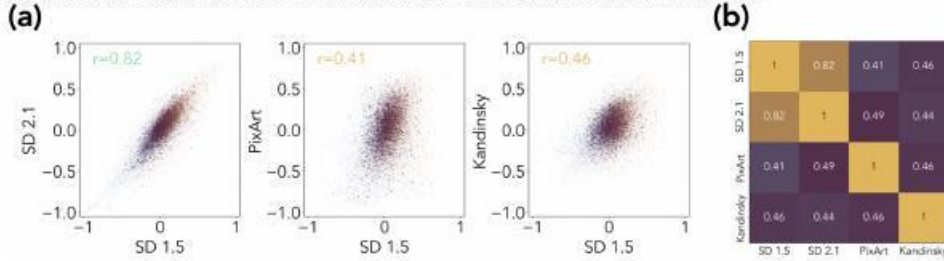


Figure 8: **Cross-model concept energy correlation.** Pairwise scatter plots of $\delta(k)$ across the four evaluated models, with Pearson correlation coefficients reported at top left. The strong alignment between SD 1.5 and SD 2.1 stands in sharp contrast to the weaker correlations with the other architectures, quantifying the extent of model-specific blindspots. **Correlation matrix of conceptual blindspots.** Heatmap of the pairwise Pearson correlations of δ across all models, summarizing how much of each model’s blindspot profile is shared.

Taken together, this analysis establishes that blindspots occupy two distinct regimes: some are universally shared—most plausibly a consequence of dataset properties—while others are strongly model-specific, arising from idiosyncrasies of training dynamics or model capacity. This dichotomy motivates the separate treatment, in the subsequent sections, of blindspots shared across models and those unique to individual models.

5.3 Qualitative Examples of Blindspots

We now examine concrete instances of both suppressed and exaggerated blindspots to characterize the concepts that populate each regime. Fig. 9a presents a blindspot suppressed by every model: all four evaluated models fail to reproduce the concept `solid white on documents`. Although the caption references this concept explicitly, none of the generations recover the intended visual semantics, demonstrating that this region of concept space is systematically under-sampled across models. Fig. 9b, by contrast, isolates a model-specific blindspot: the concept `pan` is rendered accurately by three models yet is conspicuously absent from Kandinsky’s generations. These two cases concretely instantiate the dichotomy established in Sec. 5.2 where cross-model agreement was found to be high for some concepts and limited for others.

Conversely, Fig. 10 presents a case of exaggeration, in which the concept `shadow under animal` is over-emphasized across generations. Although shadows are individually plausible, their uniformly pronounced rendering across models—set against their subtler and more variable appearance in natural images—betrays an overactive prior. Notably, a deliberate search for concepts exaggerated by only a single model yielded no clear examples, indicating that, unlike suppression, exaggeration is approximately universal across the evaluated models.

Collectively, these examples make concrete how conceptual blindspots surface in generated outputs and confirm that our energy-based diagnostic resolves both shared and model-specific failure modes. In particular, it exposes otherwise surprising limitations—such as the consistent, cross-model failure to reproduce clear or



solid background elements like whitespaces on documents—and thereby indicates that certain failure patterns originate in architectural constraints or training-data biases that transcend the idiosyncrasies of any single model.

Informative as these aggregate analyses are, they raise a sharper question: do blindspots arise only in aggregate over many samples, or do they persist at the level of individual datapoints? Addressing this question directly probes the mechanisms behind blindspots, distinguishing prompt misinterpretation from latent memorization and identifying where the two coincide.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we examine the individual natural-generated image pairs whose $\delta(\cdot)$ values, averaged over all concepts, are largest and smallest. This datapoint-level view cleanly separates model successes from failures, the latter of which we find to arise predominantly from prompt ambiguity or memorization artifacts. Fig. 11a shows pairs with near-zero average $\delta(\cdot)$, in which the generated image is conceptually indistinguishable from the original. Qualitative inspection reveals that this agreement stems not from faithful capture of the prompt semantics but from verbatim replication of memorized templates: the recurrence of near-identical visual structures (e.g., clothing outlines or object arrangements) indicates that the model is reproducing over-frequent patterns from the training data. Fig. 11b, conversely, shows pairs among the largest $\delta(\cdot)$, signaling substantial conceptual divergence between the synthesized and natural



Figure 11: • **Datapoint-level conceptual alignment.** (a) Pairs with minimal energy difference, where the model appears to reproduce memorized training patterns. (b) Pairs with large energy difference.

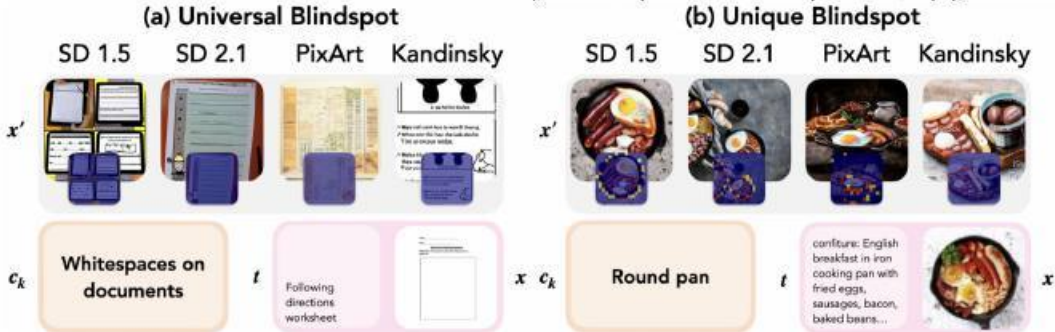


Figure 9: • **Examples of suppressed conceptual blindspots.** Each natural image x , its representative concept c_k , and its caption t are shown alongside the four synthesized images x' produced by S_θ . The universal blindspot manifests in all evaluated models, whereas the model-specific blindspot manifests only in Kandinsky.

394 images. Some of these discrepancies trace to
395 underspecified or noisy captions, but others ex-
396 pose genuine blindspots: the prompt describes
397 a clear concept that is faithfully present in $\mathcal{D}_{\mathcal{X}}$,
398 yet the model fails to realize it in $\mathcal{D}'_{\mathcal{X}}$. Such
399 cases demonstrate that even when language grounding is adequate, certain concepts lie beyond the
400 model’s generative reach.

5.5 Analyzing Post-Training Effects

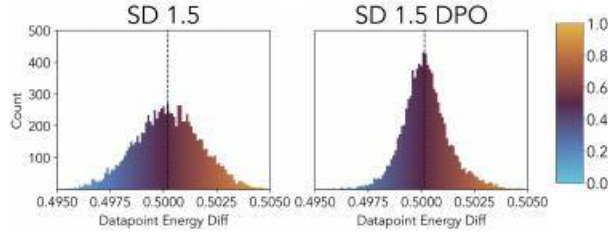


Figure 12: **Effect of DPO on concept fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distributions for SD 1.5 with and without DPO. The DPO variant shifts mass toward smaller differences, evidencing improved conceptual alignment with the natural distribution.

Post-training protocols such as safety fine-tuning have been argued to reduce the diversity of model generations [140]. Because our pipeline isolates precisely the differences between a model’s generations and the ground-truth DGP, it provides a natural instrument for quantifying the effect of DPO—a widely used safety fine-tuning protocol [141]. We compare two variants of SD 1.5, one trained with DPO and one without. For each image pair $(\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X})$ we compute the ℓ_2 distance between their concept energy vectors, $\|\xi(\mathcal{D}'_\mathcal{X}) - \xi(\mathcal{D}_\mathcal{X})\|_2$. Fig. 12 presents the resulting histogram of datapoint-wise energy differences. The DPO variant attains both a lower median and a narrower spread, demonstrating that its generated concepts align more consistently with the ground-truth DGP. DPO thus acts to regularize the model’s concept distribution, steering outputs toward the semantic content of the observed inputs. While our analysis does not disentangle the specific inductive biases DPO introduces, it furnishes direct empirical evidence that an objective favoring human-preferred generations also, and indirectly, improves agreement with the training distribution. Crucially, this improvement is bidirectional: DPO attenuates *both* the over- and the under-activation of individual concepts relative to the natural baseline. These findings underscore the value of our pipeline for characterizing post-training interventions—not merely through output quality, but through the reshaping of the conceptual geometry of the model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

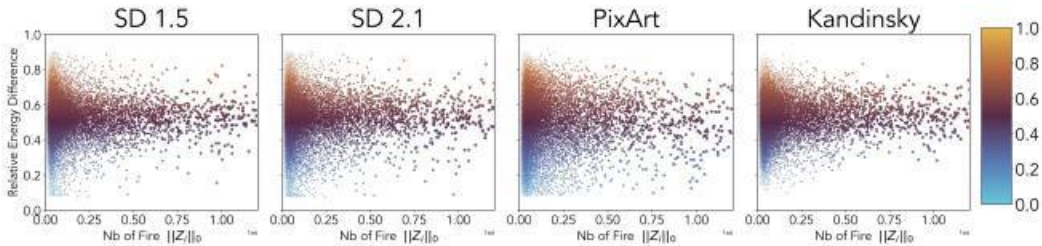


Figure 13: **Concept fidelity across the frequency spectrum.** Scatter plots relating concept frequency (x-axis) to energy difference (y-axis) for the four evaluated models. Each point is a concept, with its size proportional to its activation frequency; the trend confirms that low-frequency, long-tail concepts concentrate the largest fidelity errors.

We hypothesized above that conceptual blindspots are not solely architectural artifacts but may also arise directly from distributional properties of individual features. Here we test this hypothesis empirically, asking whether concepts that are rarely activated in natural images—those in the long tail of the data distribution—systematically induce blindspots in generative models. We process the natural dataset $\mathcal{D}_\mathcal{X}$ through the trained SAE and, for each concept k , compute its empirical frequency $\|\mathbf{Z}_{:,k}\|_0$, the count of non-zero activations of concept i across all images. We then relate this frequency to the absolute energy difference observed in the generated outputs. Fig. 13 visualizes this relationship for all four models. A clear and consistent trend emerges: high-frequency concepts exhibit small energy discrepancies, whereas rare concepts—particularly suppressed ones ($\delta(k) < 0.5$)—incur substantial alignment errors. This establishes that many blindspots are not random model quirks but

the systematic downstream consequence of long-tailed concept distributions, and it indicates that mitigating them will require not only architectural changes but also data-level interventions such as reweighting or augmentation.

6 Discussion

Our analysis uncovers numerous conceptual blindspots across four widely used generative image models. These results, moreover, only begin to exploit the framework’s reach: each individual finding could sustain a dedicated investigation of its own. Rather than pursue any single thread in depth, we have demonstrated the breadth of our method and exploratory tool, which together support—out of the box—the systematic identification of concepts a model struggles to generate, the detection of memorization artifacts, the discovery of datapoints with insufficient captions, the quantification of post-training effects, and the characterization of conceptual shifts across model architectures. This breadth positions the framework as a foundation on which follow-up work can build, whether by deepening the analysis, widening the set of evaluated architectures, or investigating the root causes of conceptual blindspots. A particularly promising direction is to endow the RA-SAE with hierarchical concept representations, enabling a correspondingly more nuanced, compositional analysis.

Limitations. Several limitations delimit the scope of our conclusions. Because concept extraction and representation rest on DINOv2 and the RA-SAE, our analysis is confined to the concepts these models encode: concepts they represent poorly will necessarily escape detection. Furthermore, although a sample of 10,000 images is substantial—and, as our concentration analysis shows, sufficient for tight estimation of δ —it may not exhaust the long tail of rare concepts, nor fully resolve concept co-occurrence and other higher-order compositional statistics.

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Appendix

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856

Table of Contents

857

A Experimental Setup	24
-----------------------------	-----------

858

A.1 Observation Space	24
-----------------------	----

859

A.2 Synthesized Images	24
------------------------	----

860

A.3  Distribution Level Analysis	24
---	----

861

A.4  Datapoint Level Analysis	25
--	----

862

A.5  Co-occurrence Analysis	25
--	----

863

B Custom RA-SAE Training Details	26
---	-----------

864

C Computational Resources	27
----------------------------------	-----------

865

D Autointerpretability	28
-------------------------------	-----------

866

E Additional Results: Model-Specific Blindspots	30
--	-----------

867

F Additional Results: Higher-order Blindspots with Compositional Discrepancy	32
---	-----------

868

G Additional Results: Blindspot Stress Testing	34
---	-----------

869

G.1 Bird Feeder Blindspot in Kandinsky	34
--	----

870

G.2 Glossy DVD Disc Blindspot in SD 1.5	35
---	----

871

H Additional Examples: Synthesized Images	36
--	-----------

872

I Concentration bounds for δ	37
---	-----------

873

J Monotonicity and Calibration-Free Interpretation of δ	38
--	-----------

874

K Stability of FID Under SAE Embeddings	39
--	-----------

875

876

877

A Experimental Setup

This section specifies the experimental setup underlying our analysis of four widely used generative image models—SD 1.5/2.1, Kandinsky, and PixArt—all trained on LAION-5B or its subsets and derivatives. The code is available at a software repository.

A.1 Observation Space

We construct the observation space by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Owing to concerns over CSAM and other unsafe content, the original release is no longer available; we therefore use a substitute release of a subset of the dataset with additional filtering of unsafe content, available at <https://huggingface.co/datasets/laion/laion2B-en-research-safe>

The sampling procedure comprises three steps: (1) loading the full LAION dataset through the Hugging Face datasets library; (2) validating each candidate—confirming a well-formed URL and resource availability via an HTTP HEAD request; and (3) sampling with replacement until 10,000 valid samples are obtained. This procedure yields D_G as a set of (x, t) tuples and guarantees an observation space of accessible image-text pairs, a non-trivial requirement given that a large fraction of the original image URLs has become unavailable since release. Additional examples of synthesized images are provided in Appendix H

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , producing triplets (x, x', t) . Concretely, for each of the 10,000 image-text pairs (x, t) in D_G , we condition each generative model g_θ on the caption t to synthesize the counterpart image x' .

Synthesis follows the standard text-to-image generation pipeline of each architecture, implemented with the Hugging Face diffusers library and executed at mixed precision (fp16). All synthetic images are generated at 512×512 resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5, used in the analysis of Sec. 5.5 is identical to the baseline SD 1.5 implementation except that its UNet is replaced by the DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1 For each evaluated model, we compute the energy difference $\delta(\cdot)$ across all 32,000 concepts, applying the sigmoid transformation with temperature $T = 0.8$ during normalization. The resulting values are rendered as log-scale density histograms with 100 bins spanning $[0, 1]$.

923 **Section 5.2** We embed the complete set of 32,000 concepts into two dimensions via UMAP applied
 924 to the sparse concept codes, coloring each point by its energy difference $\delta(\cdot)$ so as to foreground both
 925 suppressed and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise
 926 Pearson correlation coefficients between the $\delta(\cdot)$ vectors of every model pair, presenting the results
 927 as both scatter plots and correlation matrices. This analysis reveals whether blindspots cluster in
 928 conceptual space and disambiguates model-specific from universal blindspot patterns.

929 **Section 5.3** We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually examine the
 930 extrema at both ends. For suppressed blindspots we select concepts with $\delta(\cdot) < 0.1$; for exaggerated
 931 blindspots, those with $\delta(\cdot) > 0.9$. The presented examples are annotated with textual concept
 932 descriptions obtained by inspecting each concept’s most activating images and its spatial attention
 933 patterns. Our ongoing efforts to automate this concept interpretation are outlined in Appendix D

934 **Section 5.5** We compare SD 1.5 with and without DPO as follows: for each image pair (x, x') ,
 935 we compute the L2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$,
 936 applying a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before
 937 averaging. This produces datapoint-wise energy differences that quantify how far each generated
 938 image departs from its natural counterpart in concept space. The differences are then visualized as
 939 overlapping histograms that contrast the two model variants.

940 **Section 5.6** For each concept c_k , we count its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$ —the number of non-
 941 zero activations across the natural dataset—and apply a sigmoid normalization with temperature
 942 $T = 0.4$ to the energy differences $\delta(\cdot)$. The relationship is displayed as a scatter plot with empirical
 943 concept frequency on the x-axis and the sigmoid-transformed energy difference on the y-axis; point
 944 size is proportional to activation frequency and point color to the magnitude of the energy difference.

945 A.4 • Datapoint Level Analysis

946 **Section 5.4** For each image pair (x, x') , we compute the L2 norm of the difference between
 947 their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, yielding a scalar measure of conceptual divergence
 948 per pair. Ranking the samples by this divergence exposes the two failure regimes at its extremes:
 949 minimal divergence flags potential memorization artifacts, while maximal divergence flags substantial
 950 conceptual failures. The ranking thereby enables targeted qualitative inspection of specific failure
 951 modes.

952 A.5 • Co-occurrence Analysis

953 **Appendix F** For both the natural and synthesized data, D_G and D_{g_θ} , we analyze concept co-
 954 occurrence through the co-activation matrix $Z^T Z$, which encodes the pairwise correlations in concept
 955 usage. We then examine the dominant conceptual directions by eigendecomposition and assess the
 956 alignment between the natural and synthetic co-occurrence structures via cosine-similarity heatmaps
 957 between the top-100 eigenvectors of the two matrices. Each such 100×100 similarity matrix is
 958 displayed as a square heatmap: perfect diagonal structure would signify identical principal concept
 959 axes, whereas off-diagonal mass reveals rotations and mismatches in compositional geometry.

960 B Custom RA-SAE Training Details

961 This section documents the training details of our custom archetypal SAE (RA-SAE) [56], open-
962 sourced at a software repository.

963 **Dataset.** The autoencoder is trained on the complete ImageNet-1k training split (≈ 1.28 M RGB
964 images). Each image is encoded into 261 visual tokens by DINOv2 [135], and these tokens are
965 fed to the SAE without class or position embeddings. The training corpus therefore comprises
966 $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$ tokens.

967 **Dictionary.** The dictionary spans 32,000 concept dimensions. Sparsity is enforced by top- k masking
968 with $k = 5$: for each input, all activations outside the five largest are set to 0. Weights are initialized
969 via Xavier/Glorot, and training is conducted at mixed precision (fp16), with the final ten epochs
970 performed at full precision.

971 **Optimizer and schedule.** The model is trained for 50 epochs with the AdamW optimizer ($\beta_1 = 0.9$,
972 $\beta_2 = 0.999$) and weight decay 10^{-5} . The learning rate follows a linear warm-up over the first 5%
973 of steps and a subsequent cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective is an
974 MSE reconstruction loss augmented by an auxiliary term, weighted by $\lambda = 10^{-5}$, that penalizes
975 activations which never enter the top- k set.

976 C Computational Resources

977 This section itemizes the GPU resources consumed by the training and experiments reported in this
978 paper, totaling approximately 202 GPU-hours on NVIDIA H100 and H200 hardware.

979 **RA-SAE.** Training required approximately 24 GPU-hours on three NVIDIA H100s.

980 **Synthesized images.** Generating the full D_{go} (see Appendix A) took roughly 5 hours per gen-
981 erator when distributed across four NVIDIA H200 GPUs; across five generators this amounted to
982 approximately 100 GPU-hours on the H200.

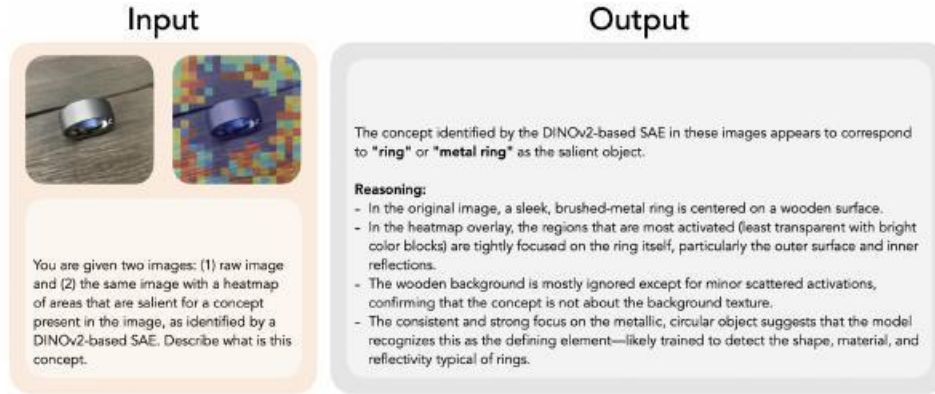
983 **• Distribution-level analysis.** Extracting distribution-level energy differences took about 3 hours
984 per generator on a single NVIDIA H200 GPU (~ 15 GPU-hours in total).

985 **• Datapoint-level analysis.** Computing datapoint-level energy differences, ranking concepts per
986 datapoint, and ranking datapoints per concept likewise took approximately 3 hours per generator on
987 one H200 GPU (~ 15 GPU-hours in total).

988 D Autointerpretability

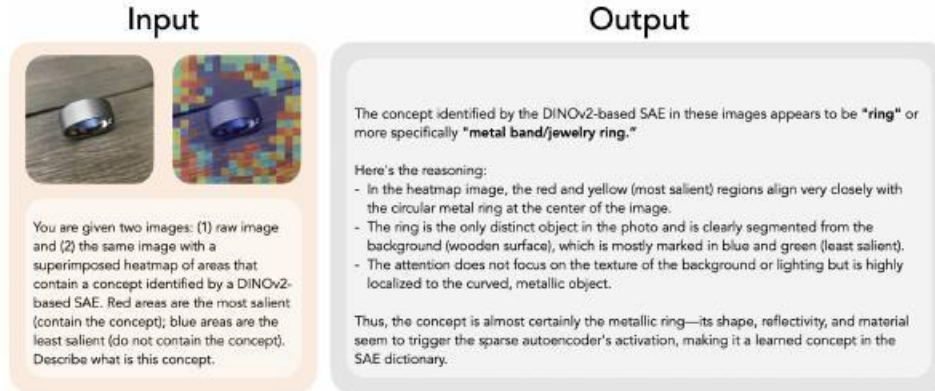
989 Toward automating the interpretation of the concepts identified by the RA-SAE, we conducted a
 990 qualitative evaluation of the ability of Vision-Language Models (VLMs) to describe them. Specifically,
 991 we queried ChatGPT 4o through the OpenAI API using a range of prompts and image formats, each
 992 derived from an image x'' in which the target concept was salient, and recorded the VLM’s description.
 993 This section reports the findings of that exploration. For evaluation we permitted the VLM to emit
 994 its full reasoning; a scalable deployment would suppress this reasoning through a minimal prompt
 995 modification requesting the description alone.

996 We first supplied the VLM with two images: a raw version of x'' and a version with a heatmap
 997 superimposed to localize the intensity of the concept:



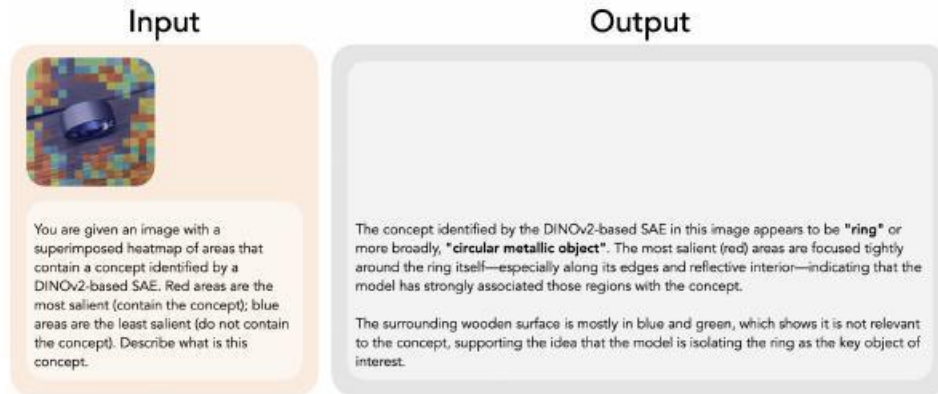
998 As the example shows, the VLM failed to interpret the heatmap correctly: rather than describing the
 999 wood texture in the background—the genuinely salient region—it described the foreground ring, to
 1000 which the RA-SAE assigned no activation.

1001 To ascertain whether this failure was an artifact of prompting, we repeated the experiment with an
 1002 expanded prompt that explained the heatmap in greater detail:



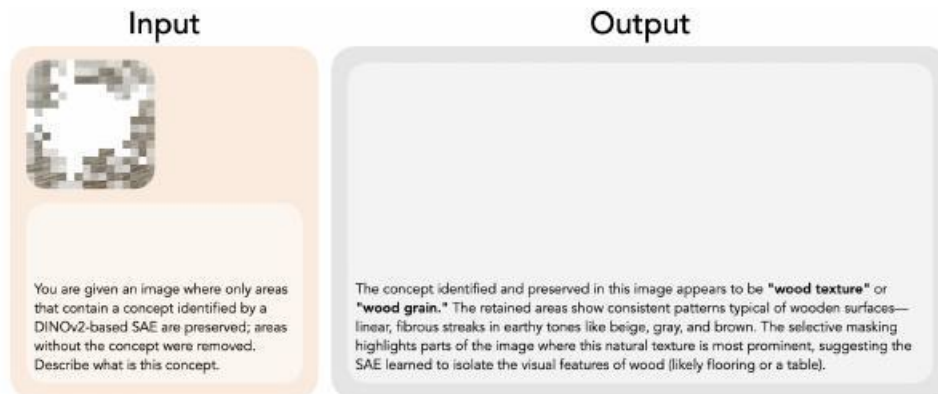
1003 This modification did not resolve the issue: the VLM again identified the ring as the prominent
 1004 concept. Several further prompting strategies—including ones that described the heatmap in still
 1005 greater depth, e.g., by supplying the code used to generate it—produced comparable results.

1006 We next provided the VLM with only the heatmap-superimposed x'' , accompanied by a similar
 1007 prompt explaining how to interpret the heatmap:



1008 Once again the VLM failed to identify the wooden background and instead focused on the ring;
 1009 varying the prompting strategy for this image format left the results largely unchanged.

1010 These observations lead us to conclude that the heatmap alone is insufficient to direct the VLM's
 1011 attention to the salient region, as the presence of less relevant regions distracts the model. To remove
 1012 this distraction, we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :



1013 As the example shows, the VLM now correctly identified wood texture as the dominant concept,
 1014 and we observed consistent success across a range of concepts and images.

1015 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
 1016 evaluation. We stress, however, that this evaluation is bounded by its qualitative nature—owing to the
 1017 absence of ground-truth annotations—and by its focus on a single VLM. We hope future work on the
 1018 autointerpretability of SAE concepts will build on and extend this analysis.

E Additional Results: Model-Specific Blindspots

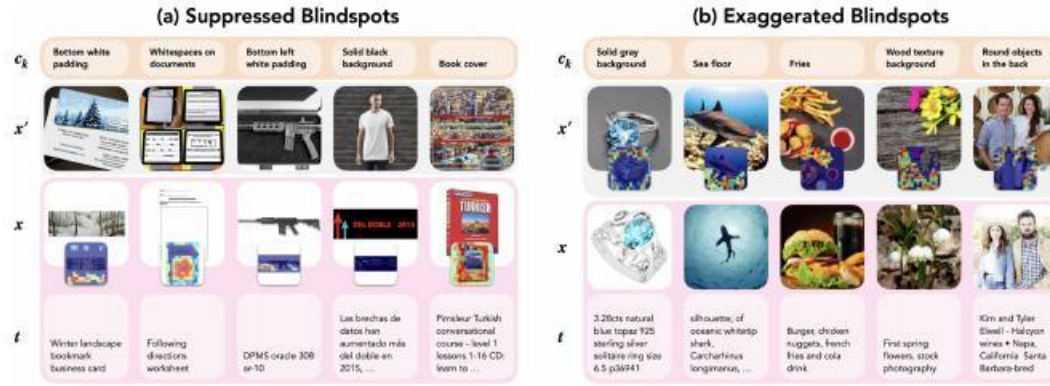


Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

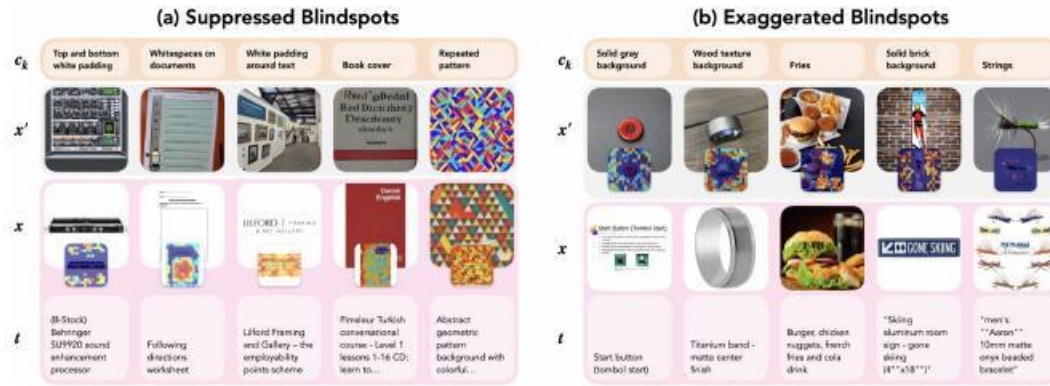


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

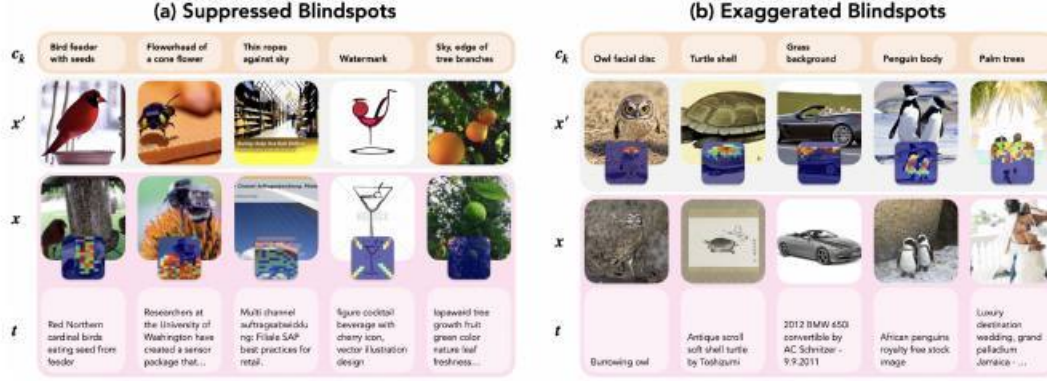


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Our analysis has so far centered on individual concept activations. Visual scenes, however, are rarely composed of isolated concepts; they are organized by rich co-occurrence patterns that encode compositional semantics. We now examine whether generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, whose entries are the pairwise correlations in concept usage.

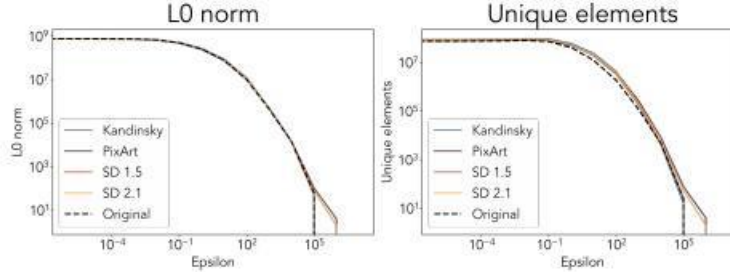


Figure 18: • **Sparsity and structural divergence.** Left: the ℓ_0 norm of the co-occurrence matrix ZZ^T as a function of the threshold ϵ , indicating how many entries remain active in each model. Right: the number of unique entries in the synthesized distribution relative to the natural distribution. Every evaluated model preserves the global sparsity structure yet diverges in activation content.

Notably, at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded across a range of ϵ values—tracks closely between the natural and synthesized distributions for every model. The gross connectivity of the conceptual graph—which concepts co-activate at all—is thus well preserved. Formally, $Z^T Z$ may be read as the adjacency matrix of a weighted, undirected graph over concepts whose edge weights encode co-activation strength across the dataset.

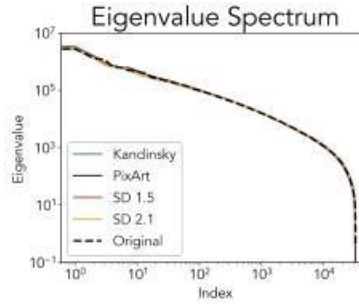


Figure 19: • **Spectral structure of co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices ZZ^T for each model and for the natural distribution. Every evaluated model reproduces the heavy-tailed decay of the natural distribution.

The specific content of these co-activations, however, diverges: as Fig. 18 (right) shows, a substantial fraction of the entries in the model-generated $Z^T Z$ are absent from the natural baseline. The capacity for compositionality is thus retained even as the identity of the active pairings shifts—a shift plausibly reflecting model-specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 presents the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. Every spectrum exhibits heavy-tailed, power-law-consistent decay, establishing that generative models preserve the overall rank structure and the allocation of variance across conceptual dimensions.

We further examine the alignment of the dominant conceptual directions via cosine-similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). Although every model exhibits partial diagonal alignment—confirming overlap in

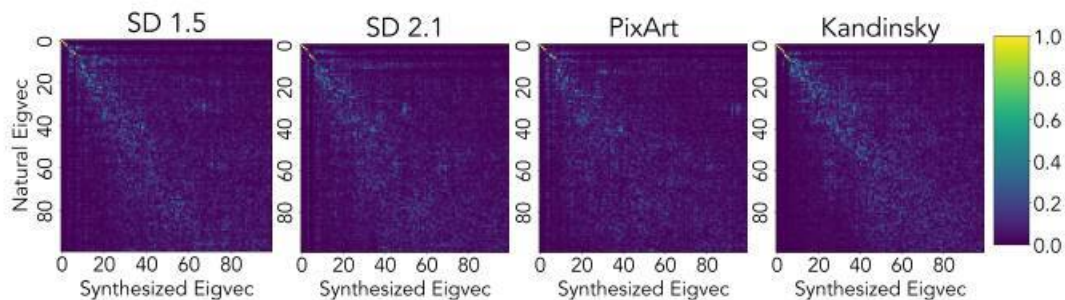


Figure 20: ● **Concept basis similarity.** Cosine-similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure indicates alignment of the dominant conceptual directions, the degree of which varies across the four models.

1045 the principal concept axes—the off-diagonal entries expose rotations and mismatches in the higher
 1046 modes, marking clear deviations in compositional geometry.

1047 Together, these findings establish a nuanced picture: diffusion models approximate the global shape
 1048 of concept co-activation remarkably well, yet deviate in subtle and structured ways once examined
 1049 through the spectral lens. Such higher-order discrepancies offer a plausible mechanism for the
 1050 observed failures in generating coherent multi-object scenes and relational concepts.

1051 G Additional Results: Blindspot Stress Testing

1052 To stress-test the blindspots identified by our method, we assembled a range of prompts describing
 1053 each blindspot, used them to generate many images, and contrasted the outputs of the models in
 1054 which the concept was flagged as a blindspot against those in which it was not.

1055 Concretely, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1056 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1057 *text-to-image model. Return these as a list of strings in Python.* We generated five images per
 1058 prompt and analyzed them with our custom RA-SAE model (Appendix B), which ranked them by
 1059 the intensity with which the target concept appeared; all images were then manually reviewed to
 1060 determine whether the blindspot was successfully depicted.

1061 As the following examples of suppressed concepts illustrate, although fragments of the target concept
 1062 occasionally surfaced—for instance a holder or string for the bird feeder blindspot, or a round
 1063 hole for the glossy DVD disc blindspot—the models systematically failed to render the complete
 1064 concept. This outcome corroborates our method’s assessment and thereby validates the stress test.

1065 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from a variety of prompts involving the bird feeder concept during stress testing. Our method flags this concept as a suppressed conceptual blindspot in Kandinsky, and the generations bear this out: the model cannot render it. SD 1.5, for which the concept was not flagged as a blindspot, generates it successfully.



Figure 22: Images generated from a variety of prompts involving the `glossy DVD disc` concept during stress testing. Our method flags this concept as a suppressed conceptual blindspot in SD 1.5, and the generations bear this out: the model cannot render it. Kandinsky, for which the concept was not flagged as a blindspot, generates it successfully.

1067 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pairs from LAION-5B, each shown with the images synthesized from the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1068 I Concentration bounds for δ

1069 In our experiments we estimate $\delta(k)$ from $n = 10,000$ paired samples per concept. This sample
 1070 budget is modest, which raises the natural question of whether it suffices for reliable estimation.
 1071 We answer it affirmatively by deriving a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ via
 1072 McDiarmid’s inequality [142]. The bound is tight and establishes that even a comparatively small
 1073 sample yields fast, accurate estimates of concept bias.

1074 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1075 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1076 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1077 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1078 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1079 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1080 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1081 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1082 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1083 which proves the claim. □

1084 In practice, the concept scores $\delta(k)$ are sparse: the majority concentrate near zero and only a few
 1085 reach values as large as 10. The concentration bound guarantees that even at the largest observed
 1086 biases, a sample of $n = 10,000$ yields estimates $\hat{\delta}_n(k)$ that deviate from the true value by no more
 1087 than a small ε with high probability. This rigorously justifies our sampling strategy and confirms that
 1088 accurate bias measurements are attainable from limited data.

1089 J Monotonicity and Calibration-Free Interpretation of δ

1090 Our objective when analyzing blindspots is to rank concepts by the severity of their generative bias, a
 1091 purpose the score $\delta(k)$ serves in practice. One might reasonably ask whether this score distorts the
 1092 ranking relative to more direct quantities such as the energy gap or the odds ratio. The following result
 1093 settles the question: $\delta(k)$ is a strictly increasing reparameterization of both and therefore inherits
 1094 their ordering exactly, so that ranking concepts by $\delta(k)$ requires no calibration whatsoever.

1095 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1096 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_X}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_X}[\xi_k(\mathbf{x})],$$

1097 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1098 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1099 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1100 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1101 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1102 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1103 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1104 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1105 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1106 concepts.

1107

□

1108 Ranking concepts by $\delta(\cdot)$ is therefore strictly equivalent to ranking them by the energy gap or by
 1109 the conceptual generation odds ρ_k : no calibration is necessary, and all three quantities induce the
 1110 identical total ordering over concepts.

1111 K Stability of FID Under SAE Embeddings

1112 In this section we establish a quantitative relationship between the Fréchet Inception Distance (FID)
 1113 computed in the original activation space (of dimension d) and the FID computed after applying
 1114 a possibly overcomplete SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout, we assume D has
 1115 orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1116 We begin by recalling an elementary fact: when D is orthogonal and square ($k = d$), it acts as an
 1117 isometry between A and Z , so that FID is preserved exactly. This regime is unrealistic, however,
 1118 and we therefore turn to the overcomplete case, where we bound the FID distortion by the extreme
 1119 singular values of D . We work with the Wasserstein-2 metric $\mathcal{W}_2$, exploiting the fact that FID is
 1120 precisely $\mathcal{W}_2^2$ specialized to Gaussians.

1121 For a probability measure μ on $\mathbb{R}^d$, we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) =$
 1122 $\mu(D^{-1}z)$. Let $\sigma_{\min}$ and $\sigma_{\max}$ denote the minimal and maximal singular values of D —equivalently,
 1123 the square roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1124 Empirically one typically finds $\sigma_{\min}, \sigma_{\max} \approx 1$, though the proof does not rely on this. We first
 1125 record a simple lemma for the non-overcomplete case.

1126 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
 1127 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
 1128 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1129 *Proof.* Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes
 1130 with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical.
 1131 $\square$

1132 This case is nonetheless unrealistic, since SAEs rely on overcompleteness precisely to extract
 1133 meaningful, interpretable concepts. In the overcomplete regime D is no longer orthonormal, yet it
 1134 can remain column-orthonormal; we leverage this property to bound the distortion in terms of the
 1135 extremal singular values of $D^\top D$.

1136 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
 1137 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
 1138 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1139 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1140 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
 1141 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1142 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1143 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
 1144 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1145 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1146 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1147 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1148 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1149 finishes the proof. $\square$

1150 In essence, Theorem 6 establishes that applying a column-orthogonal overcomplete SAE dictionary
 1151 distorts the Fréchet Inception Distance by no more than the square of its extremal singular values.
 1152 When $\mathbf{D}$ is nearly orthogonal—and empirically we consistently found $\sigma_{\min}, \sigma_{\max} \approx 1$ —the result
 1153 guarantees that FID measured in the SAE feature space is essentially indistinguishable from the
 1154 canonical FID. $\square$

1155